lncRNA expression patterns in *Pinus sylvestris* and *Picea abies* under abiotic stress

Characterizing condition-specific lncRNA neighbourhoods in cold and drought-stressed conifers

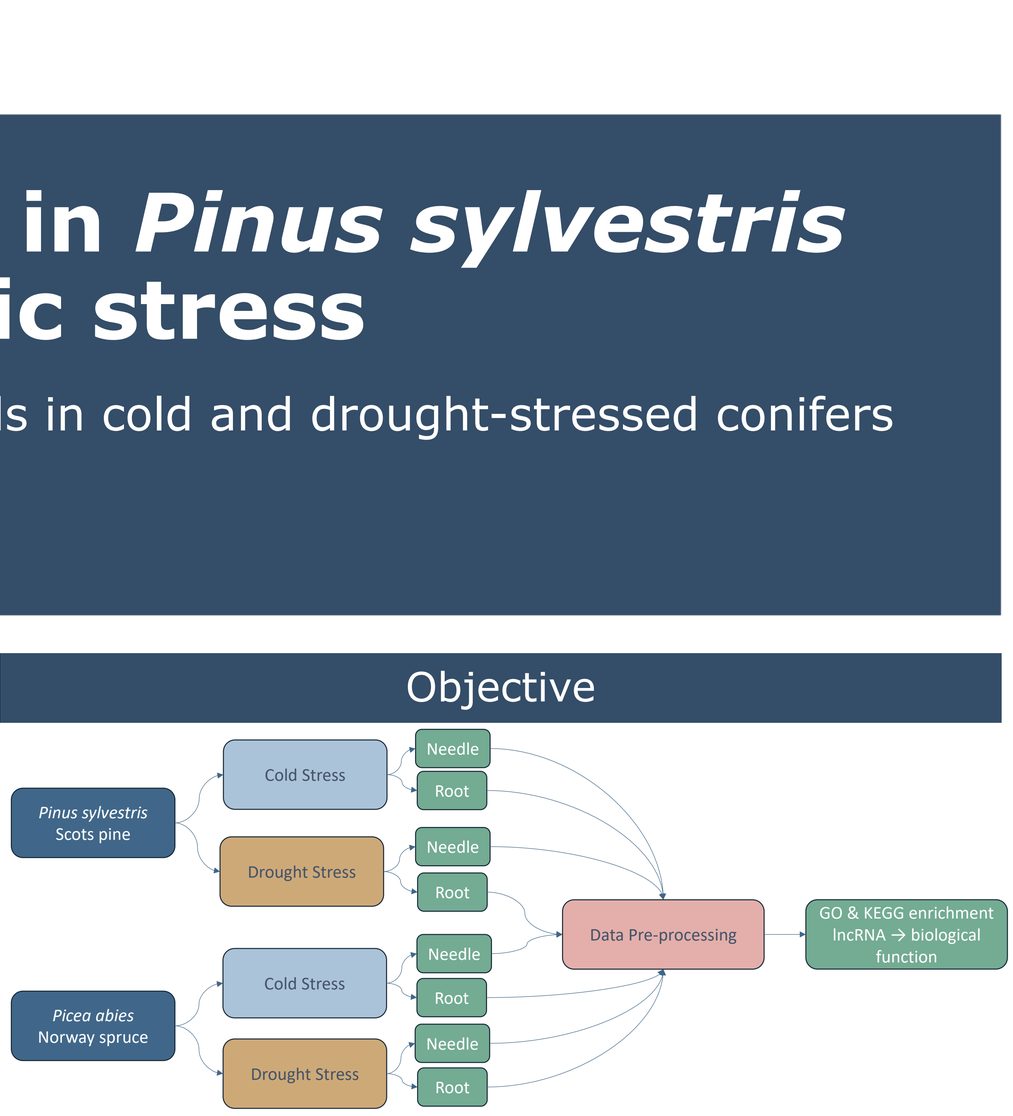
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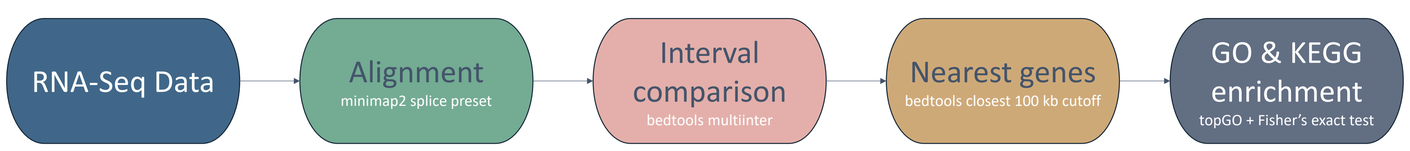
Introduction

lncRNAs are RNA transcripts longer than 200 nucleotides with no protein-coding function. In plants they regulate flowering, development, and stress responses, but in conifers they remain almost entirely unstudied. *Pinus sylvestris* and *Picea abies* are two of the most ecologically and economically important conifers in the Northern Hemisphere, yet nothing is known about how lncRNAs respond to cold or drought in either species.

Objective

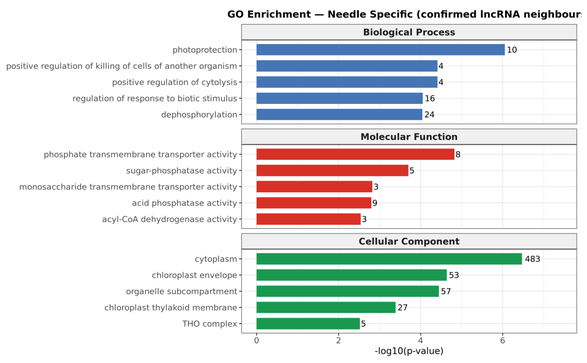


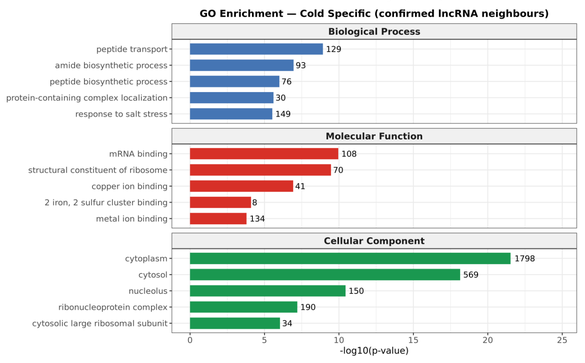
Methodology

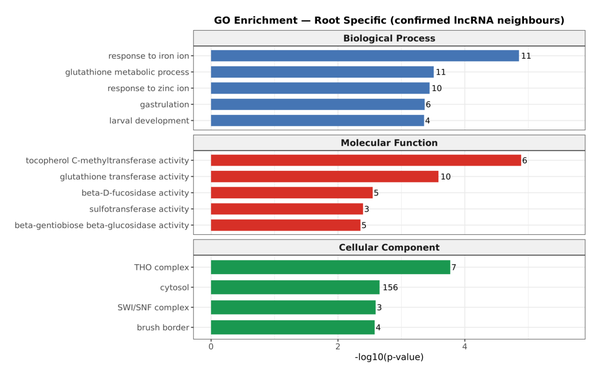


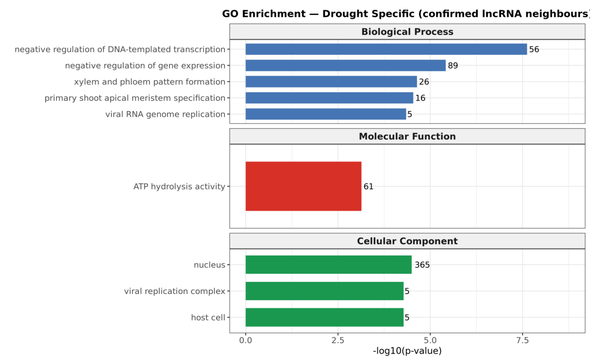
- Candidate lncRNAs identified from RNA-seq data using Plant LncRNA Pipeline v2, which filters transcripts by coding potential and length.
- Genomic intervals compared across all four samples to identify regions unique to a single tissue or stress condition.
- Protein-coding genes within 100 kb of each condition-specific region identified as functional proxies for nearby lncRNAs.
- GO and KEGG enrichment performed on neighbouring gene sets using Fisher's exact test with BH false discovery rate correction.
- Scan the **QR code** to explore the full pipeline on GitHub.

Results









- Needle-specific neighbours enriched for photoprotection and chloroplast membrane components — light management in photosynthetic tissue.
- Root-specific neighbours show iron and glutathione metabolism — antioxidant defence and metal homeostasis.
- Cold-specific regions associated with ribosomal components and mRNA binding — translational regulation as a key cold response.
- Drought-specific neighbours implicate transcriptional repression — negative regulation of gene expression is the leading signal

A note on *Picea abies*

- Spruce pipeline built against a reference transcriptome — most region IDs are reference transcript IDs, not novel transcripts.
- Confirmed lncRNA set was generated from novel transcripts — intersection is therefore very low (4–10% per category).
- Not a pipeline failure — a structural difference in how the spruce data was assembled upstream.
- Will be addressed in future work with per-sample confirmed lncRNA lists

Conclusion

- One of the first systematic characterizations of stress-responsive lncRNA expression in a conifer.
- Tissue identity and stress condition both shape lncRNA-associated gene function in *Pinus sylvestris.*
- Needle and root tissues recruit distinct biological processes — drought mobilizes a far broader response than cold.
- Spruce results preliminary due to annotation constraints — phenylpropanoid biosynthesis signal in roots is consistent with pine and warrants follow-up.
- Neighbour-based enrichment offers a reproducible framework for lncRNA functional inference in poorly annotated species.
- Pipeline fully automated and openly available for use in other conifer systems.
- Next steps: differential expression analysis and lncRNA–nearest gene co-expression to move from pattern to mechanism

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